

PATENT SPECIFICATION
UNDER 35 U.S.C. § 111(b) (PROVISIONAL APPLICATION)

**SYSTEM, METHOD, AND APPARATUS FOR ZERO-COPY IN-PLACE
DYNAMIC TOKEN ROUTING AND CAPACITY COMPACTION IN SPARSE
MIXTURE-OF-EXPERTS (MoE) ACCELERATORS**

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CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of priority under 35 U.S.C. § 119(e) of provisional patent application filing parameters established on the filing date hereof.

**STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR
DEVELOPMENT**

[0002] Not Applicable. No federal government funds or grants were used in the conception or development of this invention.

FIELD OF THE INVENTION

[0003] The present invention relates generally to deep learning hardware accelerators, parallel processors, and memory management architectures. More specifically, the present invention relates to sparse Mixture-of-Experts (MoE) neural networks, dynamic token routing, token dropping and capacity compaction, zero-copy in-place memory permuting, and high-throughput Tensor Core matrix multiplication in Large Language Models (LLMs).

BACKGROUND OF THE INVENTION

[0004] Sparse Mixture-of-Experts (MoE) architectures (e.g., Mixtral 8x7B, DeepSeek-V2/V3, Switch Transformer, GShard) achieve unprecedented parameter scaling while

maintaining bounded computational cost per token. In an MoE layer, a sequence of T tokens is dynamically routed across E specialized expert feed-forward networks based on softmax affinity scores computed by a gating network.

[0005] To prevent workload imbalance and bound accelerator memory consumption, each expert enforces a strict computational capacity limit $C = \text{capacity_factor} * (T * k / E)$. When the number of candidate tokens routed to an expert exceeds capacity C , excess tokens are discarded ('token dropping') or rerouted. However, current MoE computing platforms face severe hardware bottlenecks during token dispatch and compaction:

[0006] (1) High Auxiliary Allocation Overhead: Existing systems (e.g., Megatron-LM, DeepSpeed-MoE) allocate auxiliary global memory buffers of size $O(E * C * D)$ via operating system calls (cudaMalloc) and execute out-of-place gather/scatter operations, inducing severe transient memory spikes and triggering Out-Of-Memory (OOM) faults during long-context inference; (2) Tensor Fragmentation and SIMD Memory Disalignment: Non-contiguous candidate tokens distributed across the sequence buffer prevent dense vector memory streaming into Tensor Cores; (3) High Kernel Dispatch Latency: Sorting and repacking candidate tokens sequentially induces high runtime overheads that degrade inference throughput.

[0007] Therefore, there exists a critical need in the art for an apparatus, system, and GPU kernel method capable of dynamically routing, compacting, and truncating candidate tokens directly within expert memory buffers in-place (zero-copy) with strictly bounded $O(1)$ auxiliary scalar storage, guaranteed physical memory contiguity, and zero operating system memory allocation overhead.

BRIEF SUMMARY OF THE INVENTION

[0008] The present invention solves the aforementioned technical deficiencies by providing a hardware-software co-designed architecture and high-throughput GPU execution kernel for zero-copy in-place dynamic token routing and capacity compaction in sparse Mixture-of-Experts accelerators.

[0009] In one embodiment, a hardware MoE routing controller evaluates gating scores across candidate tokens and constructs an idempotent projection map $f_e: \{0, \dots, N-1\} \rightarrow \{0, \dots, N-1\}$ satisfying $f_e(f_e(x)) = f_e(x)$. High-affinity tokens already located within the target capacity region $[0, \dots, C-1]$ form invariant fixed points ($f_e(x) = x$) that do not oscillate or incur data movement.

[0010] In another embodiment, the controller partitions candidate token positions into mutually disjoint permutation cycles. The kernel is mapped across a 2D accelerator execution grid (pid_expert, pid_d_blk) , wherein disjoint tiles of the hidden embedding dimension D are permuted concurrently across Streaming Multiprocessors (SMs) using strictly $O(1)$ scalar auxiliary registers without boolean marking bitmasks.

[0011] Transposition fast-paths (2-cycles) swap excess low-affinity tokens in the active zone with high-affinity tokens in the tail zone in a single in-register pipeline. Following compaction, an active boundary pointer is set to C , presenting a physically contiguous token buffer directly to Tensor Cores for dense General Matrix Multiplications (GEMM) with 100% elimination of peak auxiliary VRAM.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0012] FIG. 1 is a high-level system hardware architecture diagram illustrating a sparse Mixture-of-Experts accelerator comprising High Bandwidth Memory (HBM) storing candidate

expert token buffers, a Hardware MoE Router and Token Consolidation Controller, and parallel Tensor Cores in accordance with an embodiment of the present invention.

[0013] FIG. 2 is a comparative memory state transition diagram showing physical token buffer layouts before and after idempotent in-place compaction, illustrating the transposition of tokens along disjoint permutation cycles into an active contiguous region and a dropped region.

[0014] FIG. 3 is a logical flowchart illustrating the execution steps of the zero-copy in-place MoE token dispatch and compaction algorithm executed with $O(1)$ auxiliary register memory.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

[0015] Referring to FIG. 1, a sparse Mixture-of-Experts (MoE) accelerator (100) (e.g., GPU, NPU, or domain-specific neural processor) includes a primary accelerator memory (102) storing candidate expert token embedding buffers (104) having shape $[E, N, D]$. An integrated Hardware MoE Router and Token Consolidation Controller (110) couples the memory (102) to Parallel Expert Tensor Cores (120).

[0016] The Controller (110) includes: (1) A Gating Network and Routing Affinity Evaluator (112) configured to compute softmax gating scores $S_e(t)$; (2) An Idempotent Capacity Projection Engine (114) configured to establish fixed points and cyclic orbits satisfying $f_e(f_e(x)) = f_e(x)$; and (3) A 2D-Parallel In-Place Token Permutation Kernel (116) configured to execute zero-copy token swaps across hidden dimension tiles using $O(1)$ scalar auxiliary registers.

[0017] Referring to FIG. 2, State A depicts an initial candidate token buffer where high-affinity tokens (Tok_0, Tok_2, Tok_4) are interspersed with low-affinity tokens (Tok_1, Tok_3,

Tok_5). The idempotent mapping identifies Tok_0 and Tok_2 as invariant fixed points. A 2-cycle transposition orbit is formed between Tok_1 in the active zone and Tok_4 in the tail zone. Direct register exchanges swap the embedding vectors between slots 1 and 4 across hidden dimension D. State B shows the resulting compacted buffer where slots [0, 1, 2] form a strictly contiguous memory layout ready for Tensor Core GEMM.

[0018] Referring to FIG. 3, at step 302 token routing begins. At step 304, capacity limit C is evaluated. At step 306, idempotent map f_e is generated. At step 308, the 2D execution grid is launched. Steps 310-312 check destination indices and transposition fast-paths. Step 314 executes vectorized in-register token swaps. Step 316 handles generalized cycle-leader search. At step 318, the active boundary pointer is truncated to capacity C, delivering 0.00 MB auxiliary VRAM compaction at step 320.

CLAIMS

What is claimed is:

[0019] Claim 1. A computer-implemented method for zero-copy in-place dynamic token routing and capacity compaction in a sparse Mixture-of-Experts (MoE) neural network accelerator, the method comprising: (a) evaluating routing affinity scores across N candidate tokens assigned to an expert buffer within primary accelerator memory; (b) generating an idempotent index projection map $f(i)$ satisfying $f(f(x)) = f(x)$ over an expert capacity threshold C, establishing invariant fixed points for retained tokens; (c) identifying a set of cycle leaders corresponding to mutually disjoint permutation cycles using strictly bounded $O(1)$ scalar auxiliary register storage without external boolean marking bitmasks; (d) executing an in-place cyclic permutation of multi-dimensional token embeddings along each disjoint permutation cycle using temporary

register storage, shifting token payloads directly into contiguous destination addresses; and (e) truncating an active buffer boundary pointer to capacity C to present a contiguous block of active tokens for dense matrix multiplication while dropping excess tokens without issuing host operating system memory allocation commands.

[0020] Claim 2. A sparse Mixture-of-Experts computing system comprising: a primary accelerator memory storing a plurality of candidate expert token embeddings within a contiguous buffer; an execution engine comprising a plurality of parallel tensor cores; and an integrated memory management controller operatively coupled to the memory and execution engine, configured to: compute gating routing scores for candidate tokens; construct an idempotent permutation map translating routing scores into contiguous target index addresses within an expert capacity limit C; dispatch a 2D-parallel in-place token compaction kernel to permute token embeddings along disjoint permutation cycles via cycle-leader validation using strictly bounded $O(1)$ auxiliary register memory; and output a physically contiguous, compacted token sequence without auxiliary memory buffer allocations.

[0021] Claim 3. The system of Claim 2, wherein the 2D-parallel in-place token compaction kernel comprises a 2-cycle transposition fast-path that detects reciprocal index pairings $f(f(i)) = i$ where $f(i) > i$ and executes direct in-register vectorized exchanges across embedding tiles of width BLOCK_D in a single execution pipeline.

[0022] Claim 4. The system of Claim 2, wherein the 2D-parallel in-place token compaction kernel is partitioned across a 2D accelerator execution grid (pid_expert, pid_d_blk), wherein pid_expert indexes independent expert candidate buffers and pid_d_blk indexes disjoint tiles of token hidden dimension D, executing in-place token compaction concurrently across multiple Streaming Multiprocessors with zero inter-expert memory locking.